

National Health and Nutrition Examination Survey

2013-2014 Data Documentation, Codebook, and Frequencies

Physical Activity Monitor - Minute (PAXMIN_H)

Data File: PAXMIN_H.xpt

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Component Description

The physical activity monitor (PAM) component was reintroduced to NHANES in 2011. NHANES participants were asked to wear the PAM for seven consecutive days to collect objective information on 24-hour movement when awake and asleep.

Traditionally, physical activity information has been collected in NHANES through questionnaires. Self-reported interview data, while helpful, have limitations including: 1) respondents' perceptions of activity intensity vary, and 2) periods of physical activity are difficult for respondents to recall and quantify. NHANES interview data for children and adolescents less than 16 years of age are usually provided by a proxy respondent and may be incomplete. Children may spend large amounts of time away from home and engage in sporadic periods of activity that are difficult for a proxy respondent to recall and quantify.

Objective measurement of physical activity with accelerometers had been implemented in NHANES 2003-2006. The PAM used in NHANES 2011-2014 measured acceleration (i.e., on the x-, y-, and z-axes) every 1/80th of a second (80 Hz). The device also measured ambient light levels every second (1 Hz). The 80 Hz accelerometer measurements and the 1 Hz ambient light measurements were summed over each minute, hour, and day for each participant. The following four datasets, including a header file and corresponding summary datasets at the minute, hour, and day level, were produced for the release.

A) PAM Header File (PAXHD_H):

This file contains one record for each eligible participant. It includes the information listed below:

- Whether the participant has PAM data available;
- Accelerometer device identifier;
- Day of the week the accelerometer measurements were started;
- The number of days the measurements were taken;
- The timestamps for the first and last data points;
- Whether the PAM was worn on the non-dominant hand; and
- The orientation of the PAM when it was placed on the wrist.

B) Summary datasets:

1. Minute Summary File (PAXMIN_H): one record per minute for each participant;
2. Hour Summary File (PAXHR_H): one record per hour for each participant;
3. Day Summary File (PAXDAY_H): one record per day for each participant;

Each of these minute/hour/day summary files contains multiple records per participant (who has at least one minute of data), and provides the information listed below:

- Acceleration measurements obtained on the x-, y-, or z-axes summarized at the minute level. These summary measures in the minute summary file (PAXMIN_H) are specified in Monitor-Independent Movement Summary (MIMS) units, which is a non-proprietary, open-source, device-independent universal summary metric developed by researchers at Northeastern University (John et al., 2019);
- Sums of the acceleration measurements obtained on all three axes (i.e., triaxial values from x-, y-, and z-axes) at the minute/hour/day level;
- Mean and standard deviation of ambient light (lux) at the minute level and sums of the means at the hour and day level;
- The day when the measurements used for the minute/hour/day summary record were taken (i.e., first day, second day...etc.);
- Day of the week the measurements used for the minute/hour/day summary record were taken;
- The starting time of the day, usually 00:00:00, except for the first day of measurement, in the day summary record;
- The starting data point number for the minute/hour/day summary record;
- The total minutes of data included in the hour/day summary record and total number of seconds in the minute summary record;
- Total number of the 80 Hz acceleration measurements included in the minute/hour/day summary record that were logged while the device was in idle sleep mode;
- Whether the measurements in the minute summary record were taken during a wake, sleep, non-wear, or unknown time period as classified by an algorithm (see Analytic Notes);
- Total number of valid minute measurements included in the hour/day summary record;
- Number of minutes of valid measurement included in the hour/day summary record that were classified as taken during a wake, sleep, non-wear, or unknown time period; and
- Data quality, including indicators identifying various conditions for invalid minute measurements.

Eligible Sample

In 2013-14, all participants aged 3 years and older were asked to wear a PAM for 7 consecutive days after they left the MEC.

Protocol and Procedure

Participants were asked to wear the PAM starting on the day of their exam in the NHANES Mobile Examination Center (MEC) and to keep wearing the PAM all day and night for seven full days (midnight to midnight) and remove it on the morning of the 9th day. The device used in NHANES was the ActiGraph model GT3X+, manufactured by ActiGraph of Pensacola, FL. This model was chosen because of its ability to obtain triaxial measure, its battery life, and its water resistance. According to the manufacturer's manual, the device is water resistant in accordance with the International Protection Rating: IPX7 and can be immersed in one meter of water for up to 30 minutes (ActiGraph, 2013). The devices were programmed to begin detecting and recording the magnitude of acceleration at 80 Hz sampling intervals at the end of the participant's MEC exam session and to stop recording data eight days later (i.e., on the 9th calendar day). The device was placed on a mesh wristband (like a watch), custom fitted for each participant's wrist and worn on the non-dominant hand, if possible. The participant was told not to do anything with the device except to wear it. If the participant needed to remove it, for any reason, he/she was instructed to put it back on the same wrist in the same orientation, as soon as possible.

A toll-free number and information materials describing the PAM were given to the participant. No activity logs, diaries, or records were kept by the participant. The PAMs were returned by mail in postage-paid padded envelopes provided to each participant.

For further details see the [PAM procedures manual](#).

Quality Assurance & Quality Control

Survey staff completed an examiner training program that covered the basic operating features of the PAM, equipment operation, subject recruitment strategies, PAM device initialization, and troubleshooting. Field performance and response rates were monitored by NCHS and contractor staff.

Once PAMs were received from participants, data were downloaded from the devices. In 2019, contractors at Northeastern University in Boston, MA under the direction of PAM component collaborators from the National Cancer Institute (NCI) and staff from the National Center for Health Statistics (NCHS) reviewed these data. The data quality review considered the properties of the raw acceleration measures, which are in units of gravity (g). Raw signals were processed to determine signal patterns that were unlikely to be a result of human movement. For example, maximum or minimum values ($\pm 6 g$) for extended periods of time, spikes in values, and impossible values (e.g., no gravitational acceleration observed, etc.). Individual 80 Hz accelerometer measurement data that met one or more of these conditions were flagged as "invalid". More detail about these flags is available at [Data Quality Flag Summary Table for the Physical Activity Monitor \(PAM\) Data Collected in NHANES 2011-2014 and NNYFS](#).

Data Processing and Editing

The header file and the minute/hour/day summary files were reviewed for outliers and unreasonable values. The criteria used for unreasonable data were based on published literature and expert judgment.

Not every eligible NHANES participant has a data file. For example, if a participant refused to wear the PAM, did not return it, or the data could not be retrieved from a damaged PAM, then the participant does not have a data file. Each participant may have up to 9 days of summary records. Per the protocol, the first and last day of data collection for each participant are partial days. For most participants, a complete data collection constitutes 193 hours. In some cases, e.g., due to battery depletion, data collection was shorter. A small proportion of participant data files include 194 hours.

The data files from the PAM were collected and downloaded in ActiGraph GT3x format. The GT3x files were converted to comma-separated value (csv) files following the [mHealth format](#) to facilitate processing the 80 Hz data. The minute/hour/day files are in the SAS format.

PAXSTS (Physical activity monitor data status)

This header file (PAXHD_H) variable indicates if the participant has PAM data available in the summary files. Only participants with at least one minute of 80 Hz PAM data downloaded from the returned device have records in the summary files and have PAXSTS coded as "1 (Yes)".

PAXSENID (Accelerometer device identifier)

This header file (PAXHD_H) variable indicates the unique serial number for the accelerometer device that the participant wore. The original serial number from manufacture was modified with the format retained. For budgetary reasons, accelerometer devices were reused after being sanitized during data collection. Participants who wore the same device would have the same device ID in their records.

PAXFDAY (Day of the week the accelerometer measurements were started)

This header file (PAXHD_H) variable indicates the day of the week when the PAM first started to record movement measures (i.e., Sun, Mon. etc.).

PAXLDAY (The last day when the measurements taken)

This header file (PAXHD_H) variable indicates the last day that the participant had data

recorded in the PAM. Each participant was asked to wear the PAM and have the measurements taken for 9 days. As mentioned above, the first and last day of data collection for each participant are partial days.

PAXFTIME (First data point timestamp)

This header file (PAXHD_H) variable indicates the time (in HH:MM:SS format) of the first data point in the GT3X file. For NHANES participants, this counter began at a prespecified time after each exam session in the exam center.

PAXETLDY (Timestamp at the end of the last day)

This header file (PAXHD_H) variable indicates the time (in HH:MM:SS format) at the end of the last day. This was computed from start time, number of samples, and sampling rate.

PAXHAND (PAM worn on non-dominant hand)

This header file (PAXHD_H) variable indicates whether the PAM was placed on the wrist of the non-dominant hand (the default placement for the PAM). The dominant hand is generally the one that person uses to write, or for young children, the hand used for "eating, drawing, or grabbing things". If a non-dominant hand could not be identified, then the default option was to place it on the left wrist.

PAXORENT (Surface orientation of the PAM when it was placed on the wrist)

This header file (PAXHD_H) variable indicates if the PAM was placed on the dorsal or palmar surface of the wrist while in the MEC. The default placement was the dorsal surface.

PAXDAYM, PAXDAYH, PAXDAYD (The day when PAM measurements for this minute/hour/day summary record were taken)

These variables indicate the day (1st, 2nd, 3rd, etc.) when the measurements used for the minute/hour/day summary record were taken. For example, a code of "2" indicating the measurements used in the summary record were taken on the 2nd day of PAM wear by the participant.

PAXDAYWM, PAXDAYWH, PAXDAYWD (Day of the week for the minute/hour/day summary record)

These variables indicate the day of the week (Monday, Tuesday, etc.) when the measurements used for the minute/hour/day summary record were taken.

PAXSSNMP, PAXSSNHP, PAXSSNDP (Starting 80 Hz sample number for the minute/hour/day summary record)

The 80 Hz sample number for the first data point used for the minute/hour/day summary record. For example, a value of "9600" in PAXSSNMP indicates the minute summary record used data starting from the 120th second (i.e., $9600 \div 80 \text{ Hz} = 120$) since the PAM began to measure acceleration for that participant. The sample number was identified based on the starting time the measurements were taken, regardless of whether it contained valid or flagged data.

PAXMSTD (Starting time of the day)

This day summary file (PAXDAY_H) variable indicates the starting time for PAM measurement in the day in HH:MM:SS format. It usually contains the value of "00:00:00", except for the first day of the data collection since the PAM was not worn at the beginning of the day. It is not necessarily the starting time of the MIMS computation, although they will nearly always be the same time.

PAXTSM, PAXTMH, PAXTMD (Total seconds/minutes with data in the minute file or in the hour/day files)

PAXTSM represents the total seconds of data included in the minute summary record, it includes all seconds in this minute with data and was computed from the number of 80 Hz samples and rounded to an integer value. PAXTMH and PAXTMD indicates the total minutes of data included in the hour/day summary record, and they were determined by dividing the total seconds of data in this hour/day by 60.

PAXAISMM, PAXAISMH, PAXAISMD (Total number of 80 Hz acceleration measurements logged during idle sleep mode)

These variables encoded the total number of the 80 Hz acceleration measurements included in the minute/hour/day summary record that were logged while the device was in idle sleep mode (ISM). The count includes all data, regardless whether it was flagged as "invalid" during quality control review. While taking measurements, the devices had their ISM on as a battery saving feature that was a default setting of the device firmware when NHANES data collection commenced. The device enters a sleep or low power state after experiencing 10 seconds of inactivity (fluctuation on the accelerometer < +/- 40 mg). After entering this low power state, the device checks once every second to determine if the unit has moved. While in sleep mode, the last measured accelerometer value is written into memory at the device's preset sample rate. Since the device is set to measure at 80 Hz, it will store the last known accelerometer reading 80 times every second. The device will then wake up and check for movement. If no movement (< +/- 40 mg) was detected, this pattern would continue. Otherwise, the unit would exit sleep mode and continue measuring in normal fashion. At very low levels of movement, this will impact the output of the device, including the output of MIMS-units. When ISM is triggered, very small amounts of movement are not measured; such small amounts of movement are useful to distinguish sleep and non-wear. This feature impacts the algorithm design and findings on sleep/wear/non-wear classification.

PAXVMH, PAXVMD (Total number of valid minutes in the hour/day)

These two variables in the hour/day summary files indicate the total number of minutes in the hour/day record with data considered valid because none of the measures within that minute was flagged during the QC review.

PAXMTSM, PAXMTSH, PAXMTSD (MIMS triaxial value for the minute/hour/day)

The MIMS-unit minute summary acceleration measurements obtained on the x-, y-, and z-axis (i.e., PAXMXM, PAXMYM, and PAXMZM) are summed up and released as the variable PAXMTSM in the minute summary file (PAXMIN_H). PAXMTSM is coded as "-0.01" if any of the summary measures for the x-, y-, or z-axis are coded as "-0.01" which indicates that the MIMS-unit could not be computed for that axis for that minute. The MIMS-unit calculation for the minute was based on a universal summary metric and does not take the "valid/invalid" QC flags into account. However, MIMS triaxial value for the hour/day is calculated as the sum of PAXMTSM only for valid minutes for the hour/day and is released as PAXMTSH/PAXMTSD in the hour/day summary file (i.e., PAXHR/PAXDAY).

PAXMXM, PAXMYM, PAXMZM (MIMS x-, y-, or z-axis specific value for the minute)

These variables are included in the minute summary file (PAXMIN_H) to indicate the acceleration measurements obtained on the x-, y-, or z-axis after being summarized in MIMS-units. These summaries include all seconds of data collected, regardless whether they were flagged as "invalid" during quality control review. A -0.01 value encoded in these variables means it was not possible to compute the MIMS-unit value (e.g., not enough data in the one-minute window). Sometimes, the MIMS-unit value may not be computed at the start of the day. This is very rare. Data from this minute are not included in the aggregation of the triaxial value of the minute (PAXMTSM) if "-0.01" was encoded for any of the axis summaries for the minute.

PAXPREDM (Predicted wake/sleep/non-wear status during the minute)

This variable in the minute summary file (PAXMIN_H) was established using a machine learning algorithm with the raw data collected during the minute. The PAM wearing status of the minute was classified as "wake", "sleep", "non-wear", or "unknown". A "wake" status indicates the participant is likely to be awake during the minute, and the "sleep" status indicates the participant is likely to be asleep based on the algorithm. The "non-wear" status indicates the algorithm suggested that the participant did not wear the PAM during this minute, and the "unknown" status indicates that the algorithm could not determine the status for this minute. Please see the Analytic Notes section for further details regarding the algorithm used in this feature.

PAXTRANM (Transition flag for wake/sleep/non-wear)

This variable in the minute summary file (PAXMIN_H) file indicates whether the two 30-second windows used to compute the wake/sleep/non-wear status for a given 60 seconds differ. The wake/sleep/non-wear algorithm operates on 30-second windows. A value of "1" indicates the two 30-second windows were not the same (i.e., a transition in classification of wear status).

occurred during this minute). For example, the first 30-second window of this minute record was classified as sleep, but the next 30-second window of this minute was classified as wake. The minute records that contain two 30-second windows with different status classifications (e.g., wake/sleep, etc.) are called transition minutes. When the minutes of wake, sleep, and non-wear were summed at the hour and day levels, transition minutes are flagged as "unknown".

PAXWWMH, PAXSWMH, PAXNWMH, PAXUMH (Valid wake/sleep/non-wear/unknown minutes in the hour)

These variables in the hour summary file (PAXHR_H) indicate the number of valid minutes in the hour by algorithm assigned PAM wearing status. PAXWWMH provides the valid wake wear minutes; PAXSWMH provides the valid sleep wear minutes; and PAXNWMH provides the valid non-wear minutes. PAXUMH is the number of valid minutes for which the algorithm could not make a determination (i.e., valid data exist, but appear ambiguous to the algorithm). The sum of values encoded in PAXWWMH, PAXSWMH, PAXNWMH, and PAXUMH equals to the value provided in PAXVMH (total number of valid minutes in the hour).

PAXWWMD, PAXSWMD, PAXNWMD, PAXUMD (Valid wake/sleep/non-wear/unknown minutes in the day)

These variables in the day summary file (PAXDAY_H) indicate the number of valid minutes in the day by algorithm assigned PAM wearing status. PAXWWMD provides the valid wake wear minutes; PAXSWMD provides the valid sleep wear minutes; and PAXNWMD provides the valid non-wear minutes. PAXUMD is the number of valid minutes for which the algorithm could not make a determination (i.e., valid data exist, but appear ambiguous to the algorithm). The sum of values encoded in PAXWWMD, PAXSWMD, PAXNWMD, and PAXUMD equals to the value provided in PAXVMD (total number of valid minutes in the day).

PAXLXMM, PAXLXSH, PAXLXSD (Ambient light measures for the minute/hour/day)

PAXLXMM is the mean ambient light measure (lux) for the minute and PAXLXSH and PAXLXSD are sums of the means (PAXLXMMs) at the hour and day level, respectively. The PAM device records lux every second (1 Hz). Only lux values from valid minutes are included in the computation. Lux values at or higher than 2,500 per second were coded as 2,500, thus the theoretical range for PAXLXMM is 0 to 2,500 for the minute, and 0 to 150,000 for an hour in PAXLXSH and 0 to 3,600,000 for a day in PAXLXSD.

PAXLXSDM (Standard deviation of the ambient light measures for the minute)

This variable in the minute summary file (PAXMIN_H) is the standard deviation of the lux recorded during the minute by the device. The device records lux at 1 Hz. Only lux values from valid minutes are included in the computation.

PAXQFM, PAXQFH, PAXQFD (Data quality flag for the minute/hour/day based on the quality control review)

PAXQFM indicates the number of data quality flags identified during the minute. Zero indicates that no flags occurred. Values >0 indicate that this minute is invalid based on the QC review. PAXQFH and PAXQFD indicates the number of data quality flags identified during the hour/day. Zero indicates that no flags occurred. Values >0 indicate that one or more minutes in that hour/day was flagged as "invalid". As mentioned above, the QC flag is not considered during the calculation of MIMS-unit summary; however, the variable could be used as an overall metric indicator for the quality of data on this hour/day. Since multiple flags can exist in a single minute, this variable does not represent the number of invalid minutes within that hour/day.

PAXFLGSM (Quality flag triggered)

This variable encodes the flag(s) triggered during quality review for measurements taken in this minute. Since multiple flags can be triggered during the same minute, values in this variable were recorded as strings. More detail about these flags is available at [Data Quality Flag Summary Table for the Physical Activity Monitor \(PAM\) Data Collected in NHANES 2011-2014 and NNYFS](#).

Analytic Notes

The NHANES examined sample weights should always be used for analyses with PAM data. Please refer to the [NHANES Analytic Guidelines](#) and the online [NHANES Tutorial](#) for further details on the use of the NHANES sample weights and other analytic issues.

The PAM component was reintroduced to NHANES in 2011-2014 and was included in the 2012 NHANES National Youth Fitness Survey (NNYFS). A total of 16,417 persons ages 3 years and older wore the PAM during these surveys and had data available for analysis: 2011-2012 NHANES participants 6 years and older (n=6,917); 2013-14 NHANES participants 3 years and older (n=7,776); and 2012 NNYFS participants 3-15 years (1,477). In addition, PAM data were collected for 3-5 years old participants in the 2012 NHANES (n=247) as well. However, due to disclosure concerns, these data are only available in the NCHS RDC. Overall, 96% of participants with data wore the PAM until the 9th day. About 2% of the participants with PAM data wore it for less than 7 days. In NHANES and the NNYFS, approximately 99% of the PAMs in each survey were placed on the non-dominant arm of the participant.

Users should note that 1/80th of a second of data is missing in each minute. This results in 18 seconds of data missing every 24-hour day. This was due to a computational error that was not detected until post-processing the data.

Data quality review and the MIMS-unit calculation were conducted independently. As a result, MIMS-units could be computed when QC flags are found for any given minute. Similarly, if the MIMS-unit summary couldn't be computed for a given minute (i.e., a -0.01 value was assigned to the MIMS-unit for that minute) this could possibly NOT trigger a QC flag. Data users can use the QC codes to exclude MIMS-unit data from analysis as needed.

An open-source algorithm was used with the 80 Hz accelerometer data to predict time periods of wake wear, sleep wear, or non-wear data, and assign a confidence value ranging from 0.0 to 1.0 to indicate the algorithm's confidence that the time periods are wake wear, sleep wear or non-wear. In most cases, the algorithm identified a category for each minute. Sometimes, the algorithm was uncertain how to categorize a period of time. This resulted in unknown time periods. This algorithm used three steps.

In the first step, which used 1.5 minute windows of data, signal features (e.g., signal power, variance of orientation, orientation change) were extracted from the raw data and a machine learning algorithm was used to predict whether each 30 seconds of data was wake wear, sleep wear, or non-wear. The algorithm attempted to clarify periods of data when the accelerometer's default Idle Sleep Mode (ISM) was operating. The ISM is a battery saving feature, but unfortunately it removes useful low motion time and frequency domain information that would have enhanced the algorithm's prediction. When ISM periods occurred in the data, the algorithm used the 10 seconds of data just before the PAM initiated the ISM as a proxy for the data automatically stored during the period of the ISM. For each segment of data identified as ISM data, the 10 seconds prior to the start of the segment (which contained some information characterizing low movement behavior with no fluctuation > 0.040 g) was copied over the ISM data (which contained no information). This should be considered as an approximation, because the 10 seconds of pre-ISM data was typically copied many times over throughout the ISM period. Testing suggested, however, that this small amount of information was valuable in differentiating waking low activity behavior, sleep, and non-wear time periods. Therefore, an algorithm optimized for this type of reversed-ISM data was used to categorize each 30-second window (using the two adjacent 10-second windows of data). Thirty-second windows were used because the algorithm's validation dataset used polysomnography for sleep detection, which was marked for sleep/wake state at 30 seconds intervals. It was necessary to perform this procedure, since the initial algorithm process (mentioned above) used 1.5-minute windows of data for all decision making, so waking low activity and sleep, or sleep and non-wear, were sometimes indistinguishable.

In the second step, minimum durations (i.e., wake wear time period <3 minutes, sleep wear period <10 minutes, and non-wear period <10 minutes) were filtered based on information from the prior and subsequent periods. Specifically, the summation of confidence values for each category from the current and adjacent periods (the periods immediately before and after the current period) was calculated. If the prior and subsequent periods had the same predicted category, and the sum of the confidence values of the prior and subsequent classes was greater than twice the sum of the confidence values of the predicted category from the current

period, then predicted category of the current period was switched to the predicted category of the adjacent periods. If the prior and subsequent categories did not have the same predicted category, then the predicted category of the current category was reassigned to the predicted category of the adjacent period with the higher confidence value.

In the third step, longer periods were considered. Orientation changes were used to potentially flip some labels from non-wear to sleep, or sleep to non-wear, for longer intervals of time. This was to ensure that the final predicted sleep periods included of commonly observed periodic changes in the orientation due to body reorientation on the bed during sleep.

References

- ActiGraph. GT3X+ and wGT3X+ Device Manual, 2013.
<http://s3.amazonaws.com/actigraphcorp.com/wp-content/uploads/2018/02/22094126/GT3X-wGT3X-Device-Manual-110315.pdf>
- John D, Tang Q, Albinali F, Intille S. An Open-Source Monitor-Independent Movement Summary for Accelerometer Data Processing, Journal for the Measurement of Physical Behavior. 2019;2(4),268–281.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 3 YEARS - 150 YEARS

PAXDAYM - Day of PAM measurements for this minute

Variable Name: PAXDAYM

SAS Label: Day of PAM measurements for this minute

English Text: The day (e.g., 1, 2, 3) when PAM measurements for this minute summary record were taken

Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	1st day of wear	3804932	3804932	
2	2nd day of wear	11167821	14972753	
3	3rd day of wear	11117277	26090030	
4	4th day of wear	11069782	37159812	
5	5th day of wear	11025325	48185137	
6	6th day of wear	10984638	59169775	
7	7th day of wear	10940806	70110581	
8	8th day of wear	10891210	81001791	
9	9th day of wear	7221688	88223479	
< blank >	Missing	0	88223479	

PAXDAYWM - Day of the week for this minute

Variable Name: PAXDAYWM

SAS Label: Day of the week for this minute

English Text: Day of the week for this minute (e.g., Mon, Tue, Wed)

Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Sunday	13727714	13727714	
2	Monday	13079608	26807322	
3	Tuesday	12526921	39334243	
4	Wednesday	12095064	51429307	
5	Thursday	12048922	63478229	
6	Friday	12069141	75547370	
7	Saturday	12676109	88223479	
< blank >	Missing	0	88223479	

PAXSSNMP - Starting sample number for minute

Variable Name: PAXSSNMP

SAS Label: Starting sample number for minute

English Text: Starting data point number for the minute summary record (from the 80hz sampled data)

Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 55339200	Range of Values	88223479	88223479	
.	Missing	0	88223479	

PAXTSM - Total seconds with data for this minute

Variable Name: PAXTSM

SAS Label: Total seconds with data for this minute

English Text: Total seconds with data for this minute

Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
3 to 60	Range of Values	88223479	88223479	
.	Missing	0	88223479	

PAXAISMM - Minute count: 80hz data idle sleep mode

Variable Name: PAXAISMM

SAS Label: Minute count: 80hz data idle sleep mode

English Text: Minute count: 80hz data in idle sleep mode

Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 4800	Range of Values	88223479	88223479	
.	Missing	0	88223479	

PAXMTSM - MIMS triaxial value for the minute

Variable Name: PAXMTSM

SAS Label: MIMS triaxial value for the minute

English Text: MIMS triaxial value for the minute

English Instructions: Value set as "-0.01" if any of the summary measures for the x-, y-, or z-axes was coded as "-0.01" (i.e., MIMS_unit could not be computed for that axes for the minute).

Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 22408.285	Range of Values	88221604	88221604	
-0.01	Value could not be computed	1875	88223479	
.	Missing	0	88223479	

PAXMXM - MIMS X-axis value for the minute

Variable Name: PAXMXM

SAS Label: MIMS X-axis value for the minute

English Text: MIMS X-axis value for the minute

English Instructions: Value set as "-0.01" if MIMS_unit could not be computed for the X-axes for that minute (e.g., not enough data in the one-minute window).

Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 645.693	Range of Values	88221604	88221604	
-0.01	Value could not be computed	1875	88223479	
.	Missing	0	88223479	

PAXMYM - MIMS Y-axis value for the minute

Variable Name: PAXMYM

SAS Label: MIMS Y-axis value for the minute

English Text: MIMS Y-axis value for the minute

English Instructions: Value set as "-0.01" if MIMS_unit could not be computed for the Y-axes for that minute (e.g., not enough data in the one-minute window).

Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 936.049	Range of Values	88221604	88221604	
-0.01	Value could not be computed	1875	88223479	
.	Missing	0	88223479	

PAXMZM - MIMS Z-axis value for the minute

Variable Name: PAXMZM

SAS Label: MIMS Z-axis value for the minute

English Text: MIMS Z-axis value for the minute

English Instructions: Value set as "-0.01" if MIMS_unit could not be computed for the Z-axes for that minute (e.g., not enough data in the one-minute window).

Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 22408.285	Range of Values	88221604	88221604	
-0.01	Value could not be computed	1875	88223479	
.	Missing	0	88223479	

PAXPREDM - Minute wake/sleep/non-wear prediction

Variable Name: PAXPREDM

SAS Label: Minute wake/sleep/non-wear prediction

English Text: Estimated wake/sleep/wear status during the minute

Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Wake wear	47112094	47112094	
2	Sleep wear	26073307	73185401	
3	Non wear	12091619	85277020	
4	Unknown	2946459	88223479	
< blank >	Missing	0	88223479	

PAXTRANM - Transition flag for wake/sleep/non-wear

Variable Name: PAXTRANM
SAS Label: Transition flag for wake/sleep/non-wear
English Text: Transition flag for wake/sleep/non-wear
Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	False	87709322	87709322	
1	True	514157	88223479	
< blank >	Missing	0	88223479	

PAXLXMM - Mean lux value for the minute

Variable Name: PAXLXMM
SAS Label: Mean lux value for the minute
English Text: Mean ambient light (lux) for the minute
Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 2500	Range of Values	88223479	88223479	
.	Missing	0	88223479	

PAXLXSDM - Std deviation: lux values in this minute

Variable Name: PAXLXSDM

SAS Label: Std deviation: lux values in this minute

English Text: Standard deviation: the ambient light measures (lux) in this minute

Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 1260.55	Range of Values	88223479	88223479	
.	Missing	0	88223479	

PAXQFM - Minute sum: data quality flag count

Variable Name: PAXQFM

SAS Label: Minute sum: data quality flag count

English Text: Minute sum: data quality flag count

Target: Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 14	Range of Values	88223479	88223479	
.	Missing	0	88223479	

PAXFLGSM - Quality flags triggered

Variable Name:	PAXFLGSM
SAS Label:	Quality flags triggered
English Text:	The flag(s) triggered during quality review for measurements taken in this minute.
English Instructions:	When multiple flags triggered during the same minute, the value was recorded as a string code, for example: "ABC", "EK", or "DHIJNO".
Target:	Both males and females 3 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
Quality flags triggered	Value was recorded	33625	33625	
A	Occurrence of spikes on x-axis	5256	38881	
B	Occurrence of spikes on y-axis	1535	40416	
C	Occurrence of spikes on z-axis	2799	43215	
D	Occurrence of maximum g_values on the x-axis	24	43239	
E	Occurrence of maximum g_values on the y-axis	3	43242	
F	Occurrence of maximum g_values on the z-axis	10	43252	
G	Occurrence of minimum g_values on the x-axis	1	43253	
H	Occurrence of minimum g_values on the y-axis	2	43255	
I	Occurrence of minimum g_values on the z-axis	831	44086	
J	Occurrence of contiguous maximum g_values on the x-axis	3	44089	
K	Occurrence of contiguous maximum g_values on the y-axis	2	44091	
L	Occurrence of contiguous maximum g_values on the z-axis	2	44093	
M	Occurrence of contiguous minimum g_values on the x-axis	13	44106	
N	Occurrence of contiguous minimum g_values on the y-axis	1	44107	
O	Occurrence of contiguous minimum g_values on the z-axis	7	44114	
P	Occurrence of contiguous impossible g_values for gravity (i.e., zero g for gravity)	150724	194838	
Q	Occurrence of contiguous adjacent zero values on the x-, y-, or z-axis	16884	211722	
R	Occurrence of contiguous adjacent identical non-zero values on the x-, y-, or z-axis	4338	216060	
S	Occurrence of spikes on the x-axis in 1-second intervals	187	216247	
T	Occurrence of spikes on the y-axis in 1-second intervals	74	216321	
U	Occurrence of spikes on the z-axis in 1-second intervals	14	216335	
V	Adjacent measures to periods of invalid data	57692	274027	
W	Interval jump in measurements on the x-axis (i.e., no measurement for the x-axis for this time period)	0	274027	
X	Interval jump in measurements on the y-axis (i.e., no measurement for the y-axis for this time period)	0	274027	

Code or Value	Value Description	Count	Cumulative	Skip to Item
Y	Interval jump in measurements on the z-axis (i.e., no measurement for the z-axis for this time period)	0	274027	
< blank >	Missing	87949452	88223479	

